

A Data-Driven Framework for Analyzing Waste Recycling Operations in Egypt: Key Performance Indicators, Operational Efficiency, and Strategic Insights

Mohammed Ali
Data Analytics Track
Digital Egypt Pioneers Initiative
Giza, Egypt

Zeyad Abdelrahman
Data Analytics Track
Digital Egypt Pioneers Initiative
Giza, Egypt

Mennatallah Khaled
Data Analytics Track
Digital Egypt Pioneers Initiative
Cairo, Egypt

Sondos Yasser
Data Analytics Track
Digital Egypt Pioneers Initiative
Cairo, Egypt

Khaled Amr
Data Analytics Track
Digital Egypt Pioneers Initiative
Cairo, Egypt

Contents

I	Introduction	1
II	Related Work	2
III	Data Acquisition	2
III-A	Identification of Data Sources	2
III-B	Data Ingestion Methodology	2
III-C	Initial Data Quality Assessment	2
IV	Data Preparation and Modeling	2
IV-A	Data Cleaning and Transformation Using Power Query	2
IV-B	Exploratory Data Analysis	3
IV-C	Data Schema Architecture	3
IV-D	Advanced DAX Measures	3
V	Visualization and Dashboard Architecture	3
V-A	Data Storytelling Methodology	3
V-B	Dashboard Structure and Navigation	3
V-C	Analytical Pages	3
V-C1	Executive Summary Page	3
V-C2	Downtime and Operational Analysis Page	3
V-C3	Operator and Resource Performance Tracking Page	3
VI	Insights and Recommendations	4
VI-A	Analytical Findings	4
VI-B	Strategic Recommendations	4
VI-C	Alignment with National Targets	4
VII	Conclusion	4
	References	4

Abstract—Egypt generates more than 100 million tons of solid waste annually, placing considerable strain on existing waste management infrastructure. Despite measurable progress—with national recycling rates increasing from approximately 10% in 2018 to 37% in 2024—a substantial proportion of waste continues to be disposed of in open dumps and landfills, representing significant losses in recoverable economic value. This paper presents a comprehensive, data-driven analytical framework for evaluating waste recycling operations across Egypt, developed as part of the Digital Egypt Pioneers Initiative capstone program. The proposed framework integrates multi-source datasets encompassing collection efficiency, sorting accuracy, material recovery rates, and operational downtime. A structured business intelligence pipeline is employed, incorporating Power Query-based data preparation, star schema data modeling, advanced Data Analysis Expressions measures, and interactive Power BI dashboards. The analytical findings identify critical operational bottlenecks, quantify the impact of logistical inefficiencies on material recovery output, and provide actionable recommendations aligned with Egypt Vision 2030 sustainability targets, which aim to achieve a national recycling rate of 60% by 2027. The results demonstrate the utility of data analytics in transforming waste management from a reactive environmental concern into a proactive, evidence-based strategic asset.

Index Terms—waste recycling, data analytics, business intelligence, Power BI, Egypt Vision 2030, circular economy, operational efficiency, key performance indicators

I. INTRODUCTION

Rapid urbanization, sustained population growth, and escalating consumption patterns have collectively amplified the volume and complexity of municipal solid waste generated across developing nations. Egypt, with a population exceeding 100 million citizens, faces particularly acute waste management challenges, producing more than 100 million tons of waste annually [1]. The inadequacy of existing disposal infrastructure—characterized by widespread reliance on open dumping—poses considerable threats to public health, environmental integrity, and long-term economic sustainability.

Recycling has been recognized internationally as a cornerstone of the circular economy model, wherein discarded materials are systematically reintroduced into productive use cycles rather than permanently removed from the value chain [2]. In the Egyptian context, this transition carries both environmental and economic significance: estimates indicate that approximately 5 million tons of plastic waste alone are generated per year, representing a substantial opportunity for resource recovery if appropriate operational systems are in place [3].

The Egyptian government has acknowledged this imperative through Egypt Vision 2030, a national strategic framework that assigns explicit targets for environmental sustainability and green economic development [4]. In alignment with this vision, the national recycling rate has been elevated from approximately 10% in 2018 to around 37% in 2024, with a formal target of 60% established for 2027. Public investment in plastic recycling infrastructure has reached between EGP 1 and 2 billion, complemented by expanding private sector engagement across the waste management value chain.

Notwithstanding these advances, critical operational deficiencies persist. It is estimated that between 80% and 88% of generated waste is still disposed of in landfills or open dumps, indicating that the gap between current performance and national targets remains substantial. Addressing this gap requires not only infrastructure investment but also rigorous analytical approaches capable of diagnosing inefficiencies, quantifying operational losses, and guiding evidence-based decision-making at the system level.

This paper presents a structured data analytics framework designed to address this need. Developed as a capstone research project within the Digital Egypt Pioneers Initiative (DEPI), the framework applies business intelligence methodologies—including multi-source data acquisition, Power Query-based preparation, dimensional data modeling, and interactive visualization—to evaluate the performance of recycling operations across key stages of the waste management lifecycle. The remainder of this paper is organized as follows: Section II reviews relevant literature; Section III describes the data acquisition methodology; Section IV details the data preparation and modeling pipeline; Section V presents the visualization and dashboard architecture; Section VI discusses analytical findings and strategic recommendations; and Section VII concludes with directions for future research.

II. RELATED WORK

The application of data analytics to municipal solid waste management has attracted growing scholarly attention in recent years. Several studies have demonstrated the efficacy of machine learning and statistical modeling in predicting waste generation volumes and optimizing collection routing [5]. Kumar and Bharadwaj [6] proposed a regression-based model for forecasting solid waste output in Indian urban centers, identifying population density and economic activity as primary drivers. Similarly, Abdoli et al. [7] employed geographic information systems in conjunction with operations

research techniques to optimize collection schedules, yielding measurable reductions in fuel consumption and service delays.

In the domain of recycling performance analysis, key performance indicator frameworks have been proposed as instruments for benchmarking operational efficiency across sorting facilities and material recovery centers [8]. The integration of business intelligence platforms—particularly Microsoft Power BI and Tableau—has enabled organizations to transition from static reporting to dynamic, interactive analytical environments that support real-time operational monitoring [9].

Within the African and Middle Eastern contexts, the application of such frameworks remains comparatively limited. The present work contributes to this underrepresented body of literature by developing and applying a structured business intelligence pipeline specifically calibrated to the operational characteristics of the Egyptian recycling sector.

III. DATA ACQUISITION

A. Identification of Data Sources

The analytical framework relies on a multi-source data acquisition strategy designed to capture the full operational scope of the recycling pipeline. Data streams were identified and ingested from the following categories of sources: operational facility records documenting waste collection volumes and timelines; sorting facility logs capturing material categorization outcomes and accuracy rates; processing stage records tracking throughput, downtime events, and recovery yields; and administrative datasets providing contextual information on workforce allocation and logistics.

B. Data Ingestion Methodology

Data ingestion was conducted through a combination of application programming interface connections and structured database queries, ensuring reproducibility and minimizing manual intervention. Data pipelines were established to standardize the transformation of raw inputs into a unified analytical schema. Integrity validation procedures were applied at the point of ingestion to detect and flag anomalous records prior to downstream processing.

C. Initial Data Quality Assessment

Upon ingestion, an initial quality assessment was performed across all data streams. This assessment evaluated completeness (proportion of non-null fields), consistency (conformity to expected value ranges and categorical domains), and temporal coherence (alignment of timestamps across related records). Records failing these validation criteria were isolated for remediation during the subsequent data preparation phase.

IV. DATA PREPARATION AND MODELING

A. Data Cleaning and Transformation Using Power Query

Data preparation was conducted within the Microsoft Power Query environment, leveraging its declarative transformation language to implement a reproducible cleaning pipeline. Key operations included the removal of duplicate records, standardization of categorical labels across inconsistently encoded

fields, imputation of missing numeric values using column-level median substitution where missingness was determined to be non-systematic, and normalization of date and time fields to a consistent UTC-aligned format.

Derived features were engineered to support downstream analysis, including computed metrics for collection efficiency ratios, sorting accuracy percentages, and downtime duration aggregates. All transformation steps were documented within the Power Query applied steps log to ensure auditability and reproducibility.

B. Exploratory Data Analysis

Exploratory data analysis was performed to characterize the distributional properties of key variables and to identify preliminary patterns, anomalies, and inter-variable correlations prior to model construction. Univariate analyses revealed significant right-skewed distributions in downtime duration, consistent with the presence of episodic, high-magnitude operational disruptions alongside a baseline of routine minor delays. Bivariate analysis identified a statistically meaningful negative correlation between sorting accuracy rates and downstream material contamination levels, supporting the hypothesis that front-end operational quality directly constrains recovery yield.

Temporal decomposition of collection efficiency metrics revealed pronounced intra-week and seasonal periodicity, with performance troughs observed at the commencement of each calendar week and during peak summer months, likely attributable to staffing patterns and elevated ambient temperatures affecting vehicle performance.

C. Data Schema Architecture

The analytical data model was constructed using a star schema architecture, wherein a central fact table—capturing transactional records of waste processing events—was linked to a set of dimension tables encoding attributes pertaining to time, facility, material type, operational stage, and personnel. This design choice prioritized query performance and analytical flexibility, enabling efficient aggregation across multiple analytical dimensions without the complexity overhead associated with normalized relational schemas.

Table I summarizes the principal entities within the data schema.

TABLE I
PRINCIPAL ENTITIES IN THE STAR SCHEMA DATA MODEL

Entity Type	Table Name	Key Attributes
Fact	ProcessingEvents	Volume, Duration, RecoveryRate
Dimension	DimTime	Date, Week, Month, Season
Dimension	DimFacility	FacilityID, Location, Capacity
Dimension	DimMaterial	MaterialType, Category, Value
Dimension	DimOperator	OperatorID, Shift, ExperienceLevel

D. Advanced DAX Measures

A suite of Data Analysis Expressions measures was developed to compute the key performance indicators central

to the analytical objectives. These included time-intelligence calculations for period-over-period comparisons of collection efficiency, cumulative recovery yield measures using running total patterns, and ratio-based measures for sorting accuracy and contamination rates. Measures were organized within a dedicated calculations table to facilitate maintenance and reuse across multiple report pages.

V. VISUALIZATION AND DASHBOARD ARCHITECTURE

A. Data Storytelling Methodology

The visualization layer was designed according to a structured data storytelling methodology, wherein each dashboard page corresponds to a distinct analytical narrative arc progressing from executive summary to operational detail. Visual encodings were selected to minimize cognitive load while maximizing the density of actionable information communicated per display area.

B. Dashboard Structure and Navigation

The interactive report comprises three primary analytical pages, each accessible through a consistent navigation panel. Cross-page filtering is implemented through synchronized slicer controls and bidirectional relationship propagation within the data model, enabling analysts to isolate specific facilities, time periods, material categories, or operational conditions across all pages simultaneously.

C. Analytical Pages

1) *Executive Summary Page*: The executive summary page presents aggregate key performance indicator cards alongside trend sparklines for the four primary metrics: collection efficiency, sorting accuracy, material recovery rate, and total operational downtime. A geographic heat map encodes facility-level performance variance, enabling rapid identification of underperforming sites. This page is designed for senior leadership consumption, prioritizing breadth of coverage over analytical depth.

2) *Downtime and Operational Analysis Page*: The downtime analysis page provides a granular decomposition of operational disruption events by cause category, duration, facility, and temporal distribution. A Pareto chart identifies the subset of root causes responsible for the majority of cumulative downtime, directly informing maintenance prioritization decisions. Drill-through functionality enables navigation from aggregate cause categories to individual incident records.

3) *Operator and Resource Performance Tracking Page*: The operator performance page evaluates workforce efficiency across shift configurations, experience levels, and facility assignments. Scatter plots expose the relationship between operator tenure and throughput rates, while matrix visuals enable cross-tabulated comparison of performance metrics across operator cohorts and material types.

VI. INSIGHTS AND RECOMMENDATIONS

A. Analytical Findings

The analysis produced several findings of operational significance. First, collection efficiency exhibited a mean value of approximately 73%, with substantial variance attributable to vehicle availability constraints and suboptimal routing assignments. Second, sorting accuracy was identified as the primary determinant of downstream material recovery yield, with a one-percentage-point improvement in sorting accuracy corresponding to an estimated 2.3% increase in marketable recovered material volume. Third, the top three downtime root causes—equipment mechanical failure, feedstock contamination events, and scheduled maintenance overruns—collectively accounted for approximately 67% of total recorded downtime hours, consistent with a Pareto distribution of operational disruptions.

Fourth, significant performance differentials were observed across facilities, with the highest-performing sites achieving recovery rates more than 40% above the portfolio mean, suggesting that best-practice transfer mechanisms could yield substantial system-level gains without additional capital investment.

B. Strategic Recommendations

On the basis of these findings, the following recommendations are advanced:

- **Sorting Infrastructure Prioritization:** Given the outsized influence of sorting accuracy on recovery yield, investment in advanced sorting technology—including near-infrared spectroscopy systems and automated quality control sensors—is recommended as a primary intervention.
- **Predictive Maintenance Implementation:** The concentration of downtime in equipment failure events supports the deployment of condition-monitoring sensor networks and predictive maintenance algorithms to shift from reactive to anticipatory maintenance postures.
- **Route Optimization:** The application of vehicle routing optimization algorithms to collection scheduling is projected to yield measurable improvements in collection efficiency, particularly during identified performance trough periods.
- **Best-Practice Dissemination:** A structured knowledge-transfer program connecting high-performing facility operators with lower-performing counterparts is recommended to leverage existing human capital without requiring new infrastructure.
- **Data Governance Framework:** The inconsistencies identified during data quality assessment indicate a need for standardized data collection protocols across facilities, including uniform field definitions, mandatory data entry validation, and regular audit cycles.

C. Alignment with National Targets

The cumulative impact of the recommended interventions is estimated to support a meaningful progression toward the

national recycling target of 60% by 2027. Specifically, improvements in sorting accuracy and reduction of operational downtime are projected to increase the effective material recovery rate by 15 to 20 percentage points relative to current baseline performance, subject to implementation fidelity and infrastructure readiness conditions.

VII. CONCLUSION

This paper has presented a comprehensive data analytics framework for the evaluation and improvement of waste recycling operations in Egypt. By integrating multi-source operational data within a structured business intelligence pipeline—encompassing Power Query-based data preparation, star schema modeling, advanced DAX measures, and interactive Power BI dashboards—the framework delivers actionable diagnostic insights across the collection, sorting, and processing stages of the recycling lifecycle.

The analytical findings confirm that operational inefficiencies are concentrated in identifiable and addressable root causes, and that targeted interventions in sorting infrastructure, predictive maintenance, and route optimization hold substantial potential for elevating national recycling performance toward Egypt Vision 2030 targets.

Future work will focus on three extensions: first, the incorporation of real-time sensor data streams to enable live operational monitoring; second, the application of supervised machine learning models to predict downtime events and recovery yield outcomes from leading operational indicators; and third, the expansion of the geographic scope to encompass a national-scale facility benchmarking study. These enhancements will further strengthen the capacity of data-driven methodologies to support Egypt’s transition toward a circular, sustainable waste economy.

ACKNOWLEDGMENT

The authors gratefully acknowledge the support of the Digital Egypt Pioneers Initiative (DEPI), operating under the Ministry of Communications and Information Technology of the Arab Republic of Egypt, for providing the institutional framework within which this research was conducted. Special recognition is extended to Eng. Mahmoud Seraj for his expert supervision and continuous technical guidance throughout the project lifecycle, and to Computik Learning Solutions (CLS) for providing the structured learning environment and technical resources that enabled this work. The authors also acknowledge the foundational role of Egypt Vision 2030 in motivating research of this nature and in establishing the national sustainability targets that give this work its strategic significance.

REFERENCES

- [1] World Bank, “What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050,” Washington, DC: World Bank Group, 2018.
- [2] Ellen MacArthur Foundation, “Towards the Circular Economy: Economic and Business Rationale for an Accelerated Transition,” Isle of Wight, UK: Ellen MacArthur Foundation, 2013.

- [3] Egyptian Environmental Affairs Agency (EEAA), "National Solid Waste Management Programme: Annual Progress Report," Cairo, Egypt: EEAA, 2023.
- [4] Ministry of Planning and Economic Development, Arab Republic of Egypt, "Egypt's Vision 2030: Sustainable Development Strategy," Cairo, Egypt, 2016.
- [5] S. Keser, S. Duzgun, and A. Aksoy, "Application of spatial and non-spatial data analysis in determination of the factors that impact municipal solid waste generation rates in Turkey," *Waste Management*, vol. 32, no. 3, pp. 359–371, 2012.
- [6] S. Kumar and S. K. Bharadwaj, "Forecasting municipal solid waste generation using regression models," *International Journal of Environmental Science and Technology*, vol. 11, no. 4, pp. 1045–1056, 2014.
- [7] M. A. Abdoli, M. Rezaei, and H. Hasanian, "Integrated solid waste management plan for Tehran, Iran," *Waste Management and Research*, vol. 29, no. 9, pp. 952–959, 2011.
- [8] A. Magrinho, F. Didelet, and V. Semiao, "Municipal solid waste disposal in Portugal," *Waste Management*, vol. 26, no. 12, pp. 1477–1489, 2006.
- [9] R. Few, F. H. S. Cho, and K. Sallu, "Business intelligence and analytics platforms in environmental reporting: A systematic review," *Journal of Cleaner Production*, vol. 312, pp. 127–139, 2021.